5	**(a)**	The **magnitude** of the electromotive force (e.m.f.) induced **in a circuit/conductor** is directly proportional to the rate of change of magnetic flux linkage experienced by it.	B1
	(b)(i)	-Since cylinder is rotating with **constant angular velocity**, the rate of change of magnetic flux linkage is constant. **By Faraday's Law**, the magnitude of the maximum induced emf/**peak voltage** for either direction is the **same.**	B1
		-There is an **increase** in magnetic flux linkage as magnet **approaches** the coil. And there is a **decrease** in magnetic flux linkage as magnet **leaves** the coil. **By Lenz's law**, emf is induced in **opposite** directions.	B1
	(b)(ii)	Time between 2 magnets passing coil is approximately 70 ms.	B1
		Time for 1 revolution = 4 x 70 = 280 ms	B1
		number of revolutions per minute = 60 / 0.280 = 214.	A1
	(b)(iii)	Max rate of change of flux linkage = peak voltage = 1.6 × 5.0 mV = 8.0×10^{-3} Wb s^{-1}	A1
	(c)	Since rate of revolution is doubled, the <u>rate of change of magnetic flux linkage is doubled.</u>	
		New waveform displayed on CRO:	
		• Narrower/sharper and higher peaks, as **peak voltage is doubled**	B1
		• Positive and negative peaks are **closer together**	B1
			B1

- **4 more sets of waveforms**, closer together as period of the waveform is halved

6 (a) (i) The spread of the electron beam to form bright rings due to diffraction B1